

TITLE-OF-THESIS

by

Author's-Name

Submitted in partial fulfilment of the requirements
for the degree of Title-of-Degree

at

Dalhousie University
Halifax, Nova Scotia
Month-of-Defence, Year-of-Defence

Dalhousie University is located in the Mi'kma'ki, the
ancestral and unceded territory of the Mi'kmaq.
We are all Treaty people.

© Copyright by Author's-Name, Year-of-Defence

DEDICATION

Dedication Goes Here (Optional)

...

PREFACE

I want to start by acknowledging that this Dalhousie thesis template is built on a slightly modified version of the `ocethesis.cls` class, maintained by Prof. Dan Kelley on his personal GitHub (github.com/dankelley/dal-ocethesis). This work branched off somewhere around 2015, and the `ocethesis` class was eventually passed from student to student until it reached me. Over the years, I’ve tweaked it to suit my own style while fixing some issues that arose due to deprecated packages, and whatever else happens to old code when you haven’t looked at it for a few months. I thank the previous contributors to this $\text{\TeX}$ class file, Dan Kelley, Vlado Keselj, Mike McAllister, Clyde Clements, Steven Matheson, Joseph Pallas, and others whom I have missed.

This class file, now renamed to `dalthesis.cls`, is hosted alongside this thesis `.tex` template on my GitHub (github.com/KyleBryenton/Dalhousie-LaTeX-Templates) to help take some of the pain out of the thesis writing process for future Dalhousie graduate students. This template is geared towards disciplines like chemistry or physics, and it adheres to Dalhousie Faculty of Graduate Studies (FGS) format requirements as of Summer 2025. An example of a completed thesis using this template can be seen here: (dalspace.library.dal.ca/items/2f33c6cd-166c-4031-bc01-ba1deaab62b7). I’ve left various packages and settings in the template that I found helpful during my writing process, complete with comments to help clarify what these packages can be used for.

This template uses a compact “List of Abbreviations and Symbols” that some theory-heavy students may find desirable. This compact template avoids having potentially one or two dozen pages dedicated to that part of the front matter. I’ve tried to make this as user-friendly as possible. The comments in that section of the `.tex` will guide you, and my full lists remain intact to serve as examples.

The bibliography is set up to use a provided `.bib` file using Biber (not Bib(La)TeX) as the backend. Some installations will automatically adjust to use Biber (such as Overleaf), but others may require you to change your compilation settings. I’ve left some citations in this template to showcase the bibliography format. The reference style expands on the “Nature” format by including additional information such as full article names, full author lists, digital online identifiers (DOIs), and back references

(adds hyperlinks to a reference pointing to each page it was cited in the manuscript). To help compile your bibliography file, I recommend the Google Scholar Button extension (on Chromium-based browsers, Firefox, and others) to generate the `bibtex` entries. You will sometimes need to add the pages and DOI fields manually, check author lists to convert unicode accents to $\text{\TeX}$, ensure the entry doesn't cap the number of authors listed and end with "and others", and manually abbreviate the name of the journal used. To consistently abbreviate journal names, I highly recommend UBC's Woodward Library Science and Engineering Journal Abbreviations tool: (woodward.library.ubc.ca/woodward/research-help/journal-abbreviations/).

The final step for submitting your thesis is to convert to PDF/A-1b format. However, this process can be quite tricky. To convert to PDF/A-1b, I used (xodo.com/pdf-to-pdfa) and to validate I used (www.pdfforge.org/online/en/validate-pdfa). If your `tikz` graphics are affected by this conversion, it may be due to lack of support for RGBA values. Convert your RGBA values to plain RGB, and ensure all fill opacity values used in your `tikz` are set to `fill opacity=1.0`. If this conversion causes compression of any graphics used in your thesis, I was able to alleviate this by converting $\text{PNG} \rightarrow \text{PDF (at } 1\times \text{ scale)} \rightarrow \text{PDF (at } 3\times \text{ scale)}$ using GIMP for the affected images.

Lastly, I made this template based on my own personal preferences, and any advice or suggestions offered here reflect that. If your advisor, committee, department, or faculty disagree with anything in this template, please defer to their input over my own. I hope you find this template useful.

TABLE OF CONTENTS

List of Tables	vi
List of Figures	vii
Abstract	viii
List of Abbreviations and Symbols Used	ix
Acknowledgements	x
Chapter 1 Introduction	1
1.1 Introduction Suggestions	1
1.2 Thesis Overview	1
Chapter 2 Theory	2
2.1 Theory Suggestions	2
Chapter 3 The title of your publication that has been inserted as a chapter in the main body of your thesis	4
3.1 Main Body Chapter Suggestions	5
Chapter 4 Conclusions	6
4.1 Dissertation Summary	6
4.2 Future Work	6
Bibliography	7
Appendix A Chapter 1 Supplement	8
Appendix B Chapter 2 Supplement	9
B.1 Common Unit Conversion Factors	9

LIST OF TABLES

B.1	Example Table: Conversion factors between common units used in electronic structure theory.	9
-----	--	---

LIST OF FIGURES

2.1	Example Figure #1: <code>tikz</code> example of Perdeu's Ladder.	3
A.1	Example Figure #2: An example using <code>includegraphics</code> . . .	8

ABSTRACT

As of 2025, Dalhousie's Faculty of Graduate Studies (FGS) Thesis Format Guidelines require the abstract to be one page. It may be single-spaced if necessary, but it must not include illustrations or footnotes. Note that abstracts archived by the library are truncated to 150 words for MSc theses and 350 words for PhD theses. A good abstract should summarize the work and frame it within a common theme, rather than discussing a series of disjointed papers. This is where you introduce the reader to that flow.

LIST OF ABBREVIATIONS AND SYMBOLS USED

Symbol	Description	Symbol	Description
<i>General symbols and notation</i>			
a	Constant (upright, non-bold)	h	Planck's constant, equal to a photon's energy divided by its frequency
a	Variable (italic, non-bold)	$\hbar$	Reduced Planck's constant, ($= \hbar/2\pi = 1$ a.u.)
$\mathbf{a}$	Vector (italic, bold)	$\mathbf{I}$	Identity matrix, a square matrix with ones on the main diagonal and zeros elsewhere
$\mathbf{a}$	Tensor (upright, bold)	i	Imaginary number, solution to $x^2 = -1$
$\hat{a}$	Operator (italic, non-bold, hat)	ϵ_0	Permittivity of free space ($4\pi\epsilon_0 = 1$ a.u.)
$\hat{\mathbf{a}}$	Unit vector (upright, non-bold, hat)	π	Pi, the ratio of a circle's circumference to its diameter ($= 3.14159265\dots$)
a'	Derivative of a	<i>Used atomic orbitals</i>	
$\frac{d}{dr}a$	Derivative of a with respect to r	s	$1s, 2s$
$\frac{\delta}{\delta r}a$	Functional derivative of a with respect to r	p	$2p_x, 2p_y, 2p_z, 3p_x, 3p_y, 3p_z$
Δa	Change in a	d	$3d_{z^2}, 3d_{xz}, 3d_{yz}, 3d_{xy}, 3d_{x^2-y^2}$
∇a	Gradient of a	f	$4f_{z^3}, 4f_{xz^2}, 4f_{yz^2}, 4f_{xyz}, 4f_{z(x^2-y^2)}, 4f_{x(x^2-3y^2)}, 4f_{y(3x^2-y^2)}$
$\nabla \cdot \mathbf{a}$	Divergence of $\mathbf{a}$	<i>Used indices & their typical groupings</i>	
$\nabla \times \mathbf{a}$	Curl of $\mathbf{a}$	$\cdot$	A, B, C
$\nabla^2 a$	Laplacian of a	$\cdot$	a, b, c
$\nabla \otimes \mathbf{a}$	Tensor product of ∇ and $\mathbf{a}$	$\cdot$	i_1, i_2, i_3, i_4
$\mathbf{a}^{-1}$	Matrix inverse of $\mathbf{a}$	$\cdot$	i, j, k
$\mathbf{a}^T$	Matrix transpose of $\mathbf{a}$	$\cdot$	k, ℓ, m
$\text{tr}[\mathbf{a}]$	Trace of matrix $\mathbf{a}$	$\cdot$	m, n
$\langle a_i $	Row vector of a_i	$\cdot$	p, q
$ a_i \rangle$	Column vector of a_i	$\cdot$	α, β, γ
$\langle a \rangle$	Expectation value of a (see Dirac notation)	$\cdot$	σ, σ'
<i>Common or fundamental constants</i>		$\cdot$	μ, ν
$\mathbf{1}$	Ones matrix, a matrix where every element is 1	For a complete example with a list for abbreviations and another for Latin and Greek mathematical symbols, uncomment the 2nd and 3rd tables in <code>thesis_template.tex</code>	
$\text{\AA}$	Angstrom, ($= 10^{-10}$ m)		
a_0	Bohr radius, the mean radius between an electron and the nucleus for the ground-state H atom ($= 1$ a.u.)		
e	Euler's number, the base of the natural logarithm ($= 2.71828182\dots$)		
e	Charge of an electron ($= 1$ a.u.)		

ACKNOWLEDGEMENTS

While the acknowledgements can be as short or long as desired, they typically follow the following structure:

- One paragraph thanking your committee members
- One paragraph thanking your supervisor(s) and advisor(s)
- One paragraph thanking your group, peers, and colleagues
- One paragraph thanking your friends, family, and support network
- And perhaps an extra paragraph at the start and/or end of your own creation

CHAPTER 1

INTRODUCTION

“Thirty-one years ago Dick Feynman told me about his ‘sum over histories’ version of quantum mechanics. ‘The electron does anything it likes,’ he said. ‘It just goes in any direction, at any speed, forward and backward in time, however it likes, and then you add up the amplitudes and it gives you the wave function.’ I said to him ‘You’re crazy.’ But he isn’t.” [1]

~ Prof. Freeman Dyson (1923–2020)

1.1 INTRODUCTION SUGGESTIONS

The introduction, theory, and conclusions sections are written in the main `.tex` file. These sections will be started for your Qualifying Report, and expanded on for your final thesis. A typical introduction should introduce readers to the research area, perhaps provide some historical context, and should establish why the field is important. The introduction chapter is typically 5–15 pages long.

1.2 THESIS OVERVIEW

Your introductory chapter normally concludes with a section dedicated to the thesis overview. Here, you should touch on each chapter (and perhaps major sections) in your thesis, briefly walking through what the reader can expect to find. Try to connect the themes into something that flows nicely.

CHAPTER 2

THEORY

“What I cannot create, I do not understand. Know how to solve every problem that has been solved.” [2]

~ Prof. Richard Feynman (1918–1988)

2.1 THEORY SUGGESTIONS

Your theory section should explain, in detail, all the prerequisite theory necessary to understand any manuscripts included in later chapters of the dissertation. Some opt to have 2–3 smaller chapters dedicated to theory/methods rather than just one longer one. The total length of the theory chapters is typically 20–50 pages, but could be more or less depending on the research area.

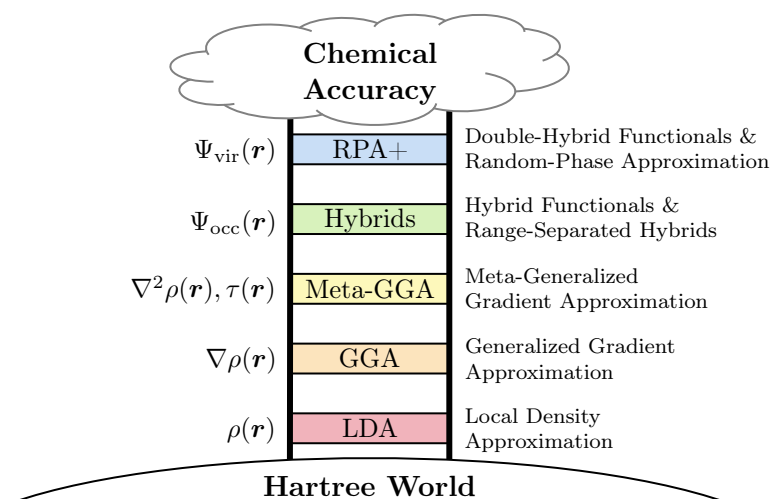


Figure 2.1: Example Figure #1: Here I include a `tikz` example of Perdew’s Ladder [3]. The `tikz` package lets you use LaTeX code to generate beautiful vector graphics using standard LaTeX sizes and fonts. This also prevents graphics from being compressed and drastically reduces the file size of your thesis. I generated many graphics in my thesis using (www.mathcha.io), and I cannot recommend it enough.

CHAPTER 3

SHORT PAPER TITLE SUITABLE FOR CHAPTER HEADING

“The career of a young theoretical physicist consists of treating the harmonic oscillator in ever-increasing levels of abstraction.” [4]

~ Prof. Sidney Coleman (1937–2007)

This chapter is adapted from: **Bryenton, K. R.**, & Johnson, E. R. *Many-Body Dispersion in Model Systems and the Sensitivity of Self-Consistent Screening*. J. Chem. Phys., **158**, 20, 204110. (2023) [5]

Author contributions: K.R.B. developed the mathematical model, wrote the software, generated all data, and wrote the first draft of the manuscript.

3.1 MAIN BODY CHAPTER SUGGESTIONS

Here we show an example of including a published paper into your thesis. I suggest making a “cover page” with the title of the article and a list of author contributions. Then you can copy-and-paste the `.tex` file (using your latest preprint, rather than the gallery proof) into your thesis. I also suggest not including the introductions from each paper, as this should already be covered in your theory chapters. Instead, include new information only and edit the initial sections of the paper to flow nicely with the previous chapters of the thesis.

In many cases, your article titles will be too long to elegantly display. Here, we effectively use three titles for this chapter: `\chapter[TOC Display]{Chapter Display} \chaptermark{Header Display}`. The “TOC Display” is used to control what this chapter is called in your Table of Contents, and I suggest using the full title of your article. “Chapter Display” is used after the “Chapter 3” heading on the previous page, and “Header Display” is used in the header of the same page. I prefer to set “Chapter Display” and “Header Display” to the same thing, and use shorter versions of the full article title.

CHAPTER 4

CONCLUSIONS

“One never notices what has been done; one can only see what remains to be done.” [6]

~ Prof. Marie Curie (1867–1934)

4.1 DISSERTATION SUMMARY

For the summary, you can walk through each chapter, briefly summarizing the findings and connecting them to convey a common theme.

4.2 FUTURE WORK

For future work, consider trying to find 2–3 new avenues your research could progress. Each one can become its own subsection.

BIBLIOGRAPHY

1. Rioux, F. *Quantum Tutorials: Section 1.55 - Using a Mach-Zehnder Interferometer to Illustrate Feynman's Sum Over Histories Approach to Quantum Mechanics* [Accessed 2025-06-12]. 2023. <https://chem.libretexts.org/@go/page/5119> (p. 1).
2. Feynman, R. *Richard Feynman's Blackboard at Time of His Death. Photo ID 1. 1029* <https://digital.archives.caltech.edu/collections/Images/1.10-29/>. [Accessed: 2025-06-12]. 1988 (p. 2).
3. Perdew, J. P. & Schmidt, K. *Jacob's ladder of density functional approximations for the exchange-correlation energy* in *AIP Conf. Proc.* **577** (2001), 1–20. doi:[10.1063/1.1390175](https://doi.org/10.1063/1.1390175) (p. 3).
4. Coleman, S. *Lectures of Sidney Coleman on Quantum Field Theory* (eds Chen, B. G.-g., Derbes, D., Griffiths, D., Hill, B., Sohn, R. & Ting, Y.-S.) Foreword by David Kaiser. ISBN: 978-9814635509. doi:[10.1142/9371](https://doi.org/10.1142/9371) (World Scientific, 2019) (p. 4).
5. Bryenton, K. R. & Johnson, E. R. Many-body dispersion in model systems and the sensitivity of self-consistent screening. *J. Chem. Phys.* **158**. (Chapter 3 in this thesis), 204110. doi:[10.1063/5.0142465](https://doi.org/10.1063/5.0142465) (2023) (p. 4).
6. Ratcliffe, S. *Oxford essential quotations* doi:[10.1093/acref/9780191866692.001.0001](https://doi.org/10.1093/acref/9780191866692.001.0001) (Oxford university press, Oxford, 2014) (p. 6).
7. Rumble, J. R. *CRC Handbook of Chemistry and Physics* ISBN: 9780367712600 (CRC press, 2021) (p. 9).

APPENDIX A

CHAPTER 1 SUPPLEMENT

Content that distracts from the flow of the dissertation, or relevant material included in the ESI of your articles, is normally placed in the appendices. Here, I've broken the appendix into "supplements" for each chapter, but this is just a personal preference.

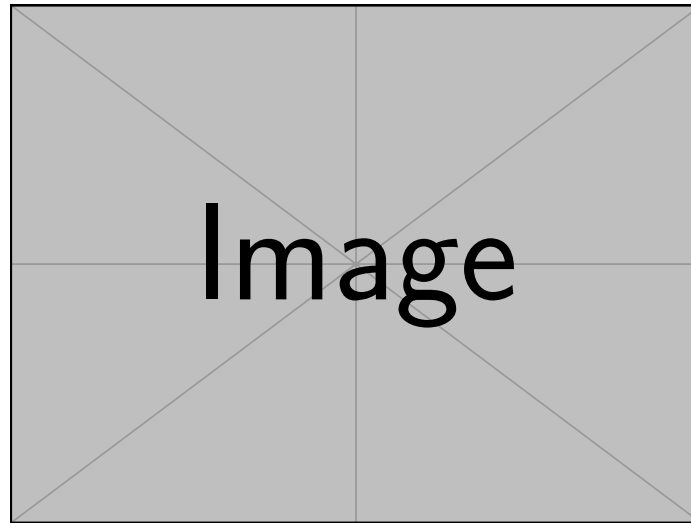


Figure A.1: Example Figure #2: Here is a figure example where we import the graphic using `includegraphics` rather than `input`.

APPENDIX B

CHAPTER 2 SUPPLEMENT

B.1 COMMON UNIT CONVERSION FACTORS

Table B.1: Example Table: Conversion factors between common energy units used in electronic structure theory to six significant figures. Units include: Hartree (Ha.), electronvolt (eV), kilocalorie per mole (kcal/mol), kilojoule per mole (kJ/mol), and wavenumber (cm^{-1}). To convert from a row unit to a column unit, multiply by the corresponding table entry. For commonly used units of length, note that $1 \text{ \AA} = 10^{-10} \text{ m} \approx 1.88973 \text{ bohr}$. Values were obtained from Ref. [7].

	Ha.	eV	kcal/mol	kJ/mol	cm^{-1}
Ha.	1	27.2114	627.510	2625.50	219475
eV	3.67493×10^{-2}	1	23.0605	96.4853	8065.54
kcal/mol	1.59360×10^{-3}	4.33641×10^{-3}	1	4.18400	349.755
kJ/mol	3.80880×10^{-4}	1.03643×10^{-3}	2.39006×10^{-1}	1	83.5935
cm^{-1}	4.55634×10^{-6}	1.23984×10^{-4}	2.85914×10^{-3}	1.19627×10^{-3}	1